1. Construct a Venn diagram to determine the validity of the given argument.

a. 1. All basketball players are well conditioned.
2. Bill is well-conditioned.
Therefore, Bill is a basketball player.

b. 1. All pilots must take a drug test.
2. Some people who take drug tests get false positives.
Therefore, some pilots get false-positives.

c. 1. All squares are rectangles.
2. Some quadrilaterals are squares.
Therefore, some quadrilaterals are rectangles.

2. Using the symbolic representations
b: I ride a bicycle to school.
s: It is snowing.
r: It is raining.
express the following statements in symbolic form.

a. It is raining and I rode my bicycle to school.

b. If it is raining or snowing, then I will not ride my bicycle to school.

3. Let x be an integer. Using the symbolic representations
p: x is a prime number.
o: x is odd.
t: x is equal to 2.
e: x is divisible by 8.
express the following in words.

a. $p \to (o \vee t)$

b. $e \to \sim p$

c. $\sim o \vee \sim e$

d. $\sim (o \wedge e) \wedge t$

4. Construct a truth table for the symbolic expression.

a. $p \wedge (q \to \sim p)$

p	q			
T	T			
T	F			
F	T			
F	F			

b. $(x \to y) \vee \sim x$

x	y			
T	T			
T	F			
F	T			
F	F			

c. $\sim a \wedge (b \to c)$

a	b	c			
T	T	T			
T	T	F			
T	F	T			
T	F	F			
F	T	T			
F	T	F			
F	F	T			
F	F	F			

5. Symbolically write the negation of the following statements.

a. The grass is not greener.

b. It is not the case that, if our car is not fixed then we cannot drive to Asheville.

6. Write the negation of the following statements.

a. Some freshmen are taking Math 107.

b. Some seniors are not taking Math 107.

c. All football players need to attend practice.

d. No professors attend football practice.

7. Using the statement, "If your order is over \$50, then you will not have to pay shipping fees." write the following:

a. Converse:

b. Inverse:

c. Contrapositive:

d. Contrapositive in terms of a sufficient condition**:**

e. Negation:

8. Using the statement, “If the sun is going down, then the cows are coming home.” write the following:

a. Converse:

b. Inverse:

c. Contrapositive:

d. Contrapositive in terms of a necessary condition**:**

e. Negation using conjunction**:**

9. Use a truth table to determine if the following arguments are valid.

a. b: You buy the bike. 1. You buy the bike or the tent.
t: You buy the tent. 2. You buy the bike.
Therefore, you do not buy the tent.

		1	2		c	Argument
b	t					
T	T					
T	F					
F	T					
F	F					

b. r: You register early.
d: You receive a discount.

1. If you register early, then you will receive a discount.
2. You register early.

Therefore, you will receive a discount.

r	d	1	2		c	Argument
T	T					
T	F					
F	T					
F	F					

c. b: You buy the property.
n: You need a loan.
p: You will pay 8%.

1. If you buy the property, then you will need a loan.
2. If you need a loan, then you will pay 8%.

Therefore, if you buy the property then you will pay 8%.

b	n	p	1	2		c	Argument
T	T	T					
T	T	F					
T	F	T					
T	F	F					
F	T	T					
F	T	F					
F	F	T					
F	F	F					

d. 1. Nothing intelligible ever puzzles me.
2. Logic puzzles me.

Therefore, logic is unintelligible.

									Argument
T	T	T							
T	T	F							
T	F	T							
T	F	F							
F	T	T							
F	T	F							
F	F	T							
F	F	F							